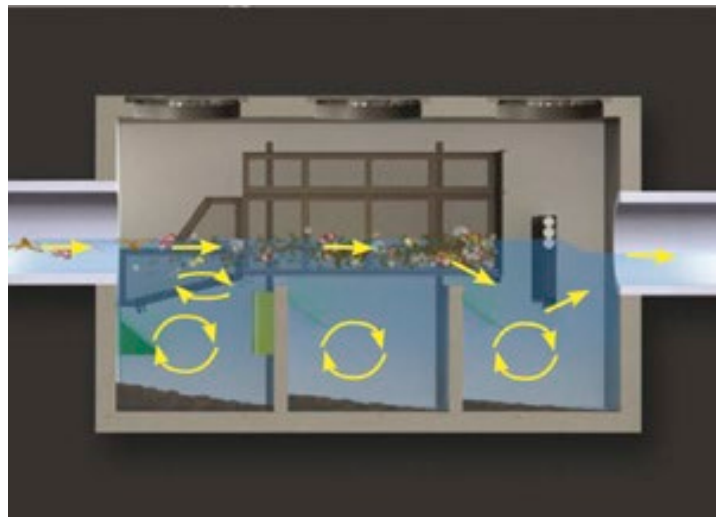




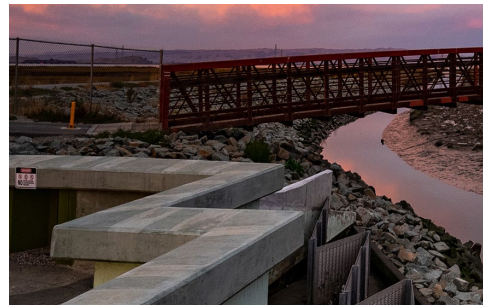
Baffle boxes are concrete or fiberglass structures containing a series of sediment settling chambers separated by baffles. The primary function of baffle boxes is to remove sediment, suspended particles, and associated pollutants from stormwater. Trash is removed through the metal cage with 5mm mesh located at the flow line of the storm drain system. Baffle boxes can be located either in-line or at the end of storm pipes.

Cleaning of baffle boxes can be accessed through a manhole or hatch above ground with a vacuum truck, with no confined spaced entry required for servicing. Screened systems are hinged for easy access to collected sediments in the baffle chambers below.



2.1.2 Louvered Panels

Louvered panels can be used in small scale and large scale applications. The most common louvered panel is the Rosco Moss Storm Flo® pictured below on the left. Louvered panels can also be used in sites that don't have a traditional catch basin and are sold by several manufacturers in just flat panel sheets of 5mm openings. The screen can be used to span across a channel of irregular shape or at the outfall of a pump station as shown below on the right in a zig zag formation to increase opening.



2.1.3 Small Scale Devices

Small scale devices include structures such as connector pipe screens and hanging baskets installed at storm drain inlets. For each large-scale device, we have prepared cost estimates for the equivalent trash capture using small scale devices. However, the small scale devices do not capture the Caltrans runoff as small-scale devices cannot be located on the major trunklines.

The connector pipe screen is a metal screen assembly installed inside a catch basin, in front of the outlet pipe, preventing debris from entering the storm drain system. They are designed to be permanently mounted to the catch basin or have a quick disconnect feature to facilitate pipe jetting or to prevent flooding should the basin drainage system become clogged. These units are designed to retain all trash and solids larger than 5 mm inside the catch basin, and once in place, it also retains large volumes of sediment. Therefore, the trash capture and storage capacity of this product are dependent upon the size of the catch basin where it is installed.



The hanging basket is similar to the connector pipe screen in that it is a separate unit that is installed into the storm drain system. Typically, installed in a catch basin, the hanging basket is an insert that keeps trash and large debris from entering the stormwater system. A frame is inserted into the top of the catch basin and is paired with a basket made with 5mm opening perforations. The large screen openings allow ample flow through and will not clog from sand and sediment loading off streets and parking lots. Inlet screens are desirable because they can be installed relatively quickly and at a relatively low cost.



3. Cost Estimates and Maintenance Considerations

3.1 Unit Costs

The following assumptions were made in determining construction costs for each proposed device:

- Installation costs for large-scale HDS and baffle devices are assumed to be 4 times the cost of the device;
- Installation of netting large-scale devices are assumed to be 3 times the cost of the device;
- Installation of louvered panels was calculated on a case-by-case basis as they are custom installations;
- Hydrodynamic separators and baffle boxes capital cost is estimated at \$3000 per treatment flow rate (cfs) based on recent bids;
- Netting and large screen device capital costs are estimated at \$2500 per treatment flow rate (cfs) based on recent bids;
- Small scale devices (i.e. inlet screens and hanging baskets) were estimated at \$2,500 per device, based on recent quotes;
- The devices are located in areas which do not require major utility relocation, property or easement acquisition or environmental permitting;
- Large scale devices were assumed to have a lifetime of 50-years;
- Small scale devices were assumed to have a lifetime of 25-years;
- Total Base Cost is presented in the 2022 construction year dollars and includes installation, construction needed to update drainage infrastructure to support trash capture device, and the device. It does not include maintenance or device replacement;
- 20% construction contingency added to total costs
- 15% design added
- Permitting costs were estimated for each specific site location.

Table 4-1 lists the unit costs used to develop each of the site's installation cost. Attachment 3 contains details of each site quantities and total cost.



Table 3-1: Unit Cost Assumptions

Item	Cost per Unit	Unit
HDS or Baffle	\$3,000	per cfs
Netting Device	\$2,500	per cfs
Louvered Panel	\$500	SF
Connector Pipe Screen	\$2,500	each
Permitting for Underground	\$27,000	lump sum
Permitting for in channel	\$105,000 - \$120,000	lump sum

3.2 Maintenance Considerations

Cleaning and maintenance costs are not included in the installation costs. They are provided in this section for reference only for the MCSTOPPP consideration. Maintenance of both small-and large-scale devices can be done by municipal staff or hired out to a cleaning contractor. In the Bay Area, there is a mix of jurisdictions that perform maintenance in-house and those that contract out the maintenance.

For cleaning, replacement, and maintenance, the following considerations should be taken:

- Large scale HDS or baffle box devices are cleaned approximately three (3) times a year, requiring 8 hours of a team of two staff members at \$100/hour, a vacuum truck at \$400/hour, or \$23,400/device annually;
- Netting devices are cleaned approximately three (3) times a year, requiring 8 hours of a team of two staff members at \$100/hour, a crane truck at \$5,000/half day, replacement net at \$1,500, or \$27,300/device annually;
- Louvered panels at the Rossi pump station are cleaned approximately three (3) times a year, requiring 4 hours of a team of two staff members at \$100/hour, a vacuum truck at \$400/hour, or \$7,200/device annually;
- Inlet screens are cleaned approximately three (3) times a year, requiring 4 hours of a team of two maintenance workers at \$100/hour- or \$7,200/device annually;

CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



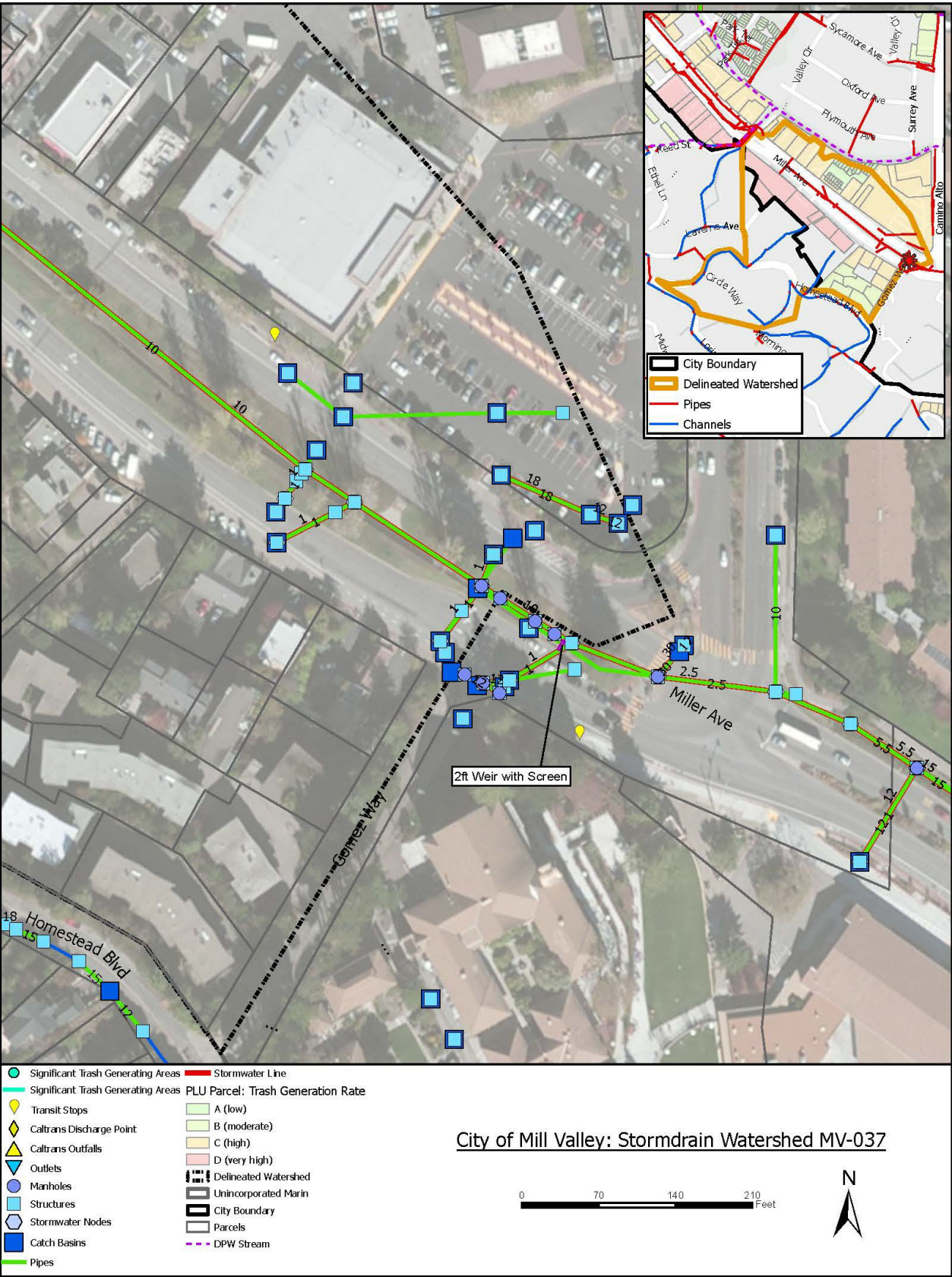
Project ID: MV-037-AIt1
Project Name: Miller Avenue
Project Location: City of Mill Valley (37.8931 , -122.5306)

Site/Device Information
Proposed Device Type: Louvered Panel Gross Solid Removal Device (GSRD) (See page 2 for Alternatives)
Project Description: Construction of a custom louvered 5-mm GSRD panel screen across the concrete channel that is 10-foot in width. The panel screen will be sited directly downstream of where the underground box culvert daylights in the median on Miller Avenue. The custom panel screen will be roughly 24 inches in height and constructed with a 6-inch weir to allow overflow to occur and avoid flooding during large storms. Once the full capture, 1-yr, 1hr design flow is exceeded, flows in the channel will overflow the screen via the weir.
Site Specific Considerations: - Location is likely tidal and known to flood - There is ample space in median for maintenance and operation of the louvered panel - Mature trees in the median would need to be avoided during construction and maintenance
O&M Summary: Device will be accessible via the turn lane on Miller Avenue.
Comments/Notes: The project may require permission from the Water Board to install a device in the channel, however it is believed the channel is not a creek and can be considered part of the City's storm drain system. The project does not receive any drainage from Caltrans' jurisdictional area.



Drainage Area and Trash Reduction Information						
Total Drainage Area: 29 acres - Mill Valley – 29 acres - Caltrans Right-of-Way – 0 acres						
Priority Land Use (PLU) Areas (acres):						
Public Agency	Trash Generation Category				Non-PLU Area	Total Area
	Low	Mod	High	Very High		
Mill Valley	1	2	5	3	18	29
Caltrans	--	--	--	--	--	0
1-yr/1-hr Peak Flow (modeled): 8 cfs						

Preliminary Cost Estimate	
Design/Permitting	\$32,000
Construction	\$12,000
Construction Contingency	\$2,400
Total Capital	\$56,000
Annual Operation & Maintenance	\$7,200/yr
Construction Cost per Acre of PLU	\$2,200/ac



State Water Resources Control Board

CERTIFIED FULL CAPTURE SYSTEM LIST OF TRASH TREATMENT CONTROL DEVICES (Updated September 2021)

Trash Provisions

In accordance with the Trash Provisions,¹ all trash treatment control devices (Devices) installed after December 2, 2015 shall meet the Full Capture System definition² and be certified by the State Water Resources Control Board (State Water Board) Executive Director, or designee, prior to installation. The Devices included on this list are either: 1) new Device applications certified by the State Water Board after adoption of the Trash Provisions, or 2) grandfathered device fact sheets listed by the San Francisco Regional Water Board prior to the adoption of the Trash Provisions.

The Trash Provisions require that only Certified Full Capture Devices may be installed, and that the Devices be designed according to the following criteria:

1. Appropriately sized to treat not less than the peak flowrate resulting from a one--year, one-hour storm event (design storm) or at least the same peak flows from the corresponding storm drain;
2. Do not bypass trash below the design storm under maximum operational loading conditions; and
3. Trap all particles that are 5 mm or greater up to the design flow³ or at least the same peak flows from the corresponding storm drain; and do not have a

¹ Amendment to the Water Quality Control Plan for Ocean Waters of California to Control Trash and Part 1 Trash Provisions of the Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, And Estuaries of California adopted by the State Water Board.

² A Full Capture System is a treatment control, or series of treatment controls, including but not limited to, a multi-benefit project or a low-impact development control that traps all particles that are 5 mm or greater, and has a design treatment capacity that is either: of not less than the peak flow rate, Q, resulting from a one-year, one-hour storm in the subdrainage area, or b) appropriately sized to, and designed to carry at least the same flows as, the corresponding storm drain.

³ The region specific one-year, one-hour storm (or design flow) may be obtained from the [National Oceanic and Atmospheric Precipitation Estimates](https://www.weather.gov/media/owp/oh/hdsc/docs/Atlas14_Volume6.pdf) (https://www.weather.gov/media/owp/oh/hdsc/docs/Atlas14_Volume6.pdf)

CERTIFIED FULL CAPTURE SYSTEM LIST OF TRASH TREATMENT CONTROL DEVICES

diversion structure present upstream such that a portion of the peak flow is not treated to trap all particles 5-millimeter or greater.

Vector Control Accessibility

According to the California Health and Safety Code⁴, California landowners are legally responsible to abate (eliminate the source of) a public nuisance arising from their property, including mosquitoes. Mosquito vector control districts have substantial authority to access public and private property, inspect known or suspected sources of mosquitoes, abate mosquito sources, and charge the landowner for work performed and/or charge fees if a landowner is unwilling or unable to address a mosquito source arising from their property.

Depending on its design, certain Devices may impede the mosquito vector control district's ability to (1) visually inspect the Device and/or storm vault for mosquito breeding, and (2) apply the appropriate chemical treatment. Moreover, some devices may create a habitat for mosquitoes. Prior to installation of any certified Device, the local mosquito vector control district should be contacted to ensure the installation conforms to the district's visual inspection, treatment, and vector breeding minimizing guidelines. The Mosquito Vector Control Association of California may also be contacted via email at Trashtreatment@mvcac.org.

New Device Application Certification or Fact Sheet Update

To apply for certification of a new Device, or to update a grandfathered Device fact sheet, the Device owner shall submit an application/fact sheet in accordance with the *Trash Treatment Control Device Certification and Fact Sheet Update Requirements*. Upon determining that a Device application is complete and meets the definition of a trash full capture system and is approved by the Mosquito Vector Control Association of California, the Executive Director will place the Device on the State Water Board's *Certified Full Capture System List of Trash Treatment Control Devices*.

The Trash Treatment Control Device Certification and Fact Sheet Update Requirements is found on the [Trash Implementation Program webpage](https://www.waterboards.ca.gov/water_issues/programs/stormwater/trash_implementation.html) (https://www.waterboards.ca.gov/water_issues/programs/stormwater/trash_implementation.html).

Certification of any Device does not constitute an endorsement by the State Water Board. The Executive Director reserves the right to de-certify and remove any Device from this list that does not satisfy the requirements of the Trash Provisions, such as but not limited to when a Device is discontinued, found to not trap trash in accordance with the Trash Provisions, or no longer has Mosquito Vector Control Association of California approval.

⁴ Health & Safety Code sections 2001- 4(d); 2002; 2060 (b) and Health & Safety Code sections 2060-2067, 100170, and 100175.